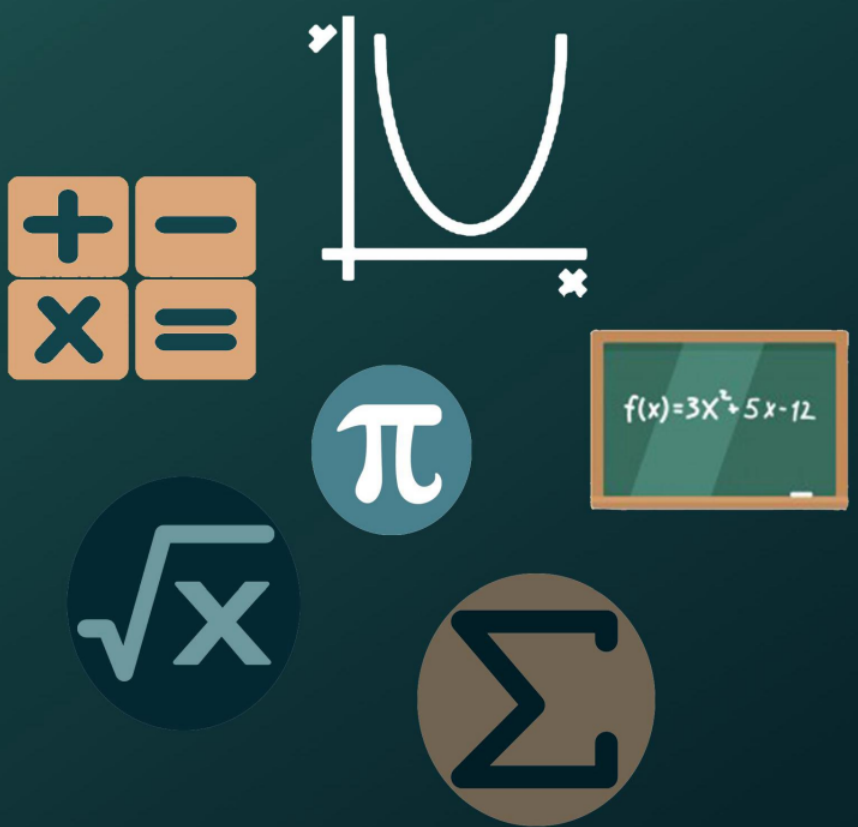
MATHEMATICS

3 - Dimensional Geometry



SHEET

eSaral.com

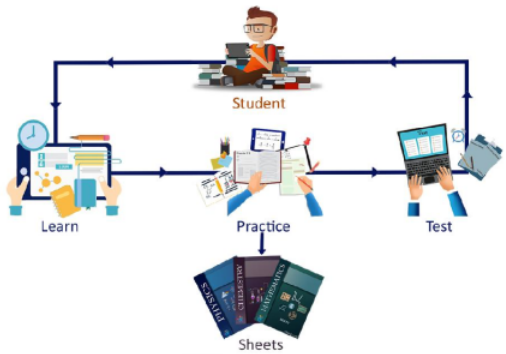
Contents

1.	Module Description	Page - ii
2.	Homework Index	Page - iii
3.	Exercise 1	Page - 1
4.	Exercise 1A	Page - 5
5.	Exercise 2	Page - 9
6.	Exercise 2A	Page - 12
7.	Answer Key	Page - 17

Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex-1	Ex-1A	Ex-2	Ex-2A
1	Distance Formula	1,2			
2	Section Formula	3,4	2,10		
3	Direction cosine of Vector/Line	5-9	12		1
4	General Equation of Plane	10-12	13	1-2	
5	Equation of a Plane Passing Through Three Given Points	13			
6	Intercept Form of the Plane	14,15		3-6	2
7	Perpendicular Distance of a Point from a Plane	16,17	11,14	7-10	
8	Angle Between Two Planes	18-20	1,15		3-5, 28-29
9	Equation of the Bisector Planes			11	6-7
10	Family of Planes	21	16,17	12	8,32
11	Two Sides of a Plane				
12	Straight Lines	22,23			
13	Problems Based on Straight lines	24-32	18-21	13-16	9-11, 33,34
14	Angle Between a Line and a Plane	33-37		17-18	12
15	General Equation of the Plane containing a Line or Lines		22-24	19-24	13-16, 30,35
16	Shortest Distance Between Two Lines	38	25-27		25
17	Unsymmetrical Form of Straight Line	39	9		17-18
18	Line of Greatest Slope in a Plane				
19	After Completing Chapter		5-7	25	19-24,26, 27,31,36

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Distance Formula

Q1. Perpendicular distance of the point (3, 4, 5) from the y-axis, is :

(A) $\sqrt{41}$ (B) 4

(C) 5 (D) $\sqrt{34}$

Q2. The projection of any line on co-ordinate to be respectivley 3, 4, 5, then its length is :

(A) 50

(B) 12

(C) $5\sqrt{2}$

(D) None of these

Section Formula

Q3. The co-ordinates of the point in which the line joining the points (3, 5, –7) and (–2, 1, 8) is intersected by the plane *yz* are given by :

(A) $\left(0,-\frac{13}{5},-2\right)$ (B) $\left(0,-\frac{13}{5},\frac{2}{5}\right)$

(C) $\left(0,\frac{13}{5},\frac{2}{5}\right)$ (D) $\left(0,\frac{13}{5},2\right)$

Q4. The co-ordinates of the point which divides the join of the points (2, –1, 3) and (4, 3, 1) in the ratio 3 : 4 internally are given by :

(A) $\frac{2}{7},\frac{20}{7},\frac{10}{7}$ (B) $\frac{10}{7},\frac{15}{7},\frac{2}{7}$

(C) $\frac{20}{7},\frac{5}{7},\frac{15}{7}$ (D) $\frac{15}{7},\frac{20}{7},\frac{3}{7}$

Direction cosine of Vector/Line

Q5. The angle between the lines whose direction cosines satisfy the equations $l+m+n=0$, $l^2+m^2-n^2=0$ is given by :

(A) $\frac{\pi}{6}$ (B) $\frac{5\pi}{6}$

(C) $\frac{\pi}{3}$ (D) $\frac{2\pi}{3}$

Q6. The number of straight line that are equally inclined to three dimensional co-ordinate axes, is :

(A) 8 (B) 6

(C) 4 (D) 2

Q7. The direction ratios of the diagonals of a cube with side length 1 cm which joins the origin to the opposite corner are (when the 3 concurrent edges of the cube are co-ordinate axes) :

(A) 1, 2, 3 (B) 2, –2, 1

(C) 1, 1, 1 (D) $\frac{2}{\sqrt{3}},\frac{2}{\sqrt{3}},\frac{2}{\sqrt{3}}$

Q8. A line makes angle of 45° and 60° with the positive axes of X and Y respectivley. The angle made by the same line with the positive axis of Z is :

(A) 60° or 120°

(B) 30° or 60°

(C) 90° or 120°

(D) 60° or 90°

Q9. If a line make the angle α, β and γ with the axes respectivley, then

$\cos 2\alpha+\cos 2\beta+\cos 2\gamma$ is equal to :

(A) 1 (B) –1

(C) 2 (D) –2

General Equation of Plane

Q10. The equation of a plane which passes through (2, –3, 1) and is normal to the line joining the points (3, 4, –1) and (2, –1, 5) is given by :

(A) $x-5y+6z-19=0$

(B) $x+5y+6z+19=0$

(C) $x-5y-6z-19=0$

(D) $x+5y-6z+19=0$

Q11. The equation of the plane through (2, 3, 4) and parallel to the plane $x+2y+4z=5$ is:

(A) $x+2y+4z=10$

(B) $x+2y+4z=3$

(C) $x+2y+4z=24$

(D) $x+y+2z=2$

Q12. The direction cosines of the normal to the plane $x+2y-3z+4=0$ are :

(A) $-\frac{1}{\sqrt{14}},-\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}$

(B) $\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}$

(C) $-\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},\frac{3}{\sqrt{14}}$

(D) $\frac{1}{\sqrt{14}},\frac{2}{\sqrt{14}},-\frac{3}{\sqrt{14}}$

Equation of a Plane Passing Through Three Given Points

Q13. Equation of the plane containing the lines.

$\vec{r}=\hat{i}+2\hat{j}-\hat{k}+\lambda(\hat{i}+2\hat{j}-\hat{k})$ and

$\vec{r}=\hat{i}+2\hat{j}-\hat{k}+\mu(\hat{i}+\hat{j}+3\hat{k})$ is :

(A) $7x-4y-z=0$

(B) $7(x-1)-4(y-1)-(z+3)=0$

(C) $x+2y-z=0$

(D) $x+y+3z=0$

Intercept Form of the Plane

Q14. The equation of a plane which cuts equal intercepts of unit length on the axes, is :

(A) $x+y+z=0$ (B) $x+y+z=1$

(C) $x+y-z=1$ (D) $\frac{x}{a}+\frac{y}{a}+\frac{z}{a}=1$

Q15. If the plane $x-3y+5z=d$, passes through the point (1, 2, 4), then the intercepts cut by it on the axes of x, y, z are respectivley :

(A) 1, –6, 20 (B) 15, –5, 3

(C) 1, –5, 3 (D) –15, 5, –3

Perpendicular Distance of a Point from a Plane

Q16. The distance of the point (2, 3, 4) from the plane $3x-6y+2z+11=0$ is :

(A) 2 (B) 9

(C) 10 (D) 1

Q17 The image of the point (–1, 3, 4) in the plane x – 2y = 0 is -

(A) (15, 11, 4) (B) $\left(-\frac{17}{3},-\frac{19}{3},1\right)$

(C) (8, 4, 4) (D) None of these

Angle Between Two Planes

Q18. If the planes $ax+by+cz+d=0$ and $a'x+b'y+c'z+d'=0$ be mutually perpendicular, then:

(A) $aa'+bb'+cc'+dd'=0$

(B) $aa'+bb'+cc'=0$

(C) $\frac{a}{a'}=\frac{b}{b'}=\frac{c}{c'}$

(D) $\frac{a}{a'}+\frac{b}{b'}+\frac{c}{c'}=0$

Q19. Distance between parallel planes
$2x-2y+z+3=0$ and
$4x-4y+2z+5=0$ is :

(A) $\frac{1}{6}$ (B) $\frac{1}{4}$

(C) $\frac{1}{2}$ (D) $\frac{1}{3}$

Q20. The angle between the planes
$2x-y+z=6$ and $x+y+2z=7$, is :

(A) 45° (B) 0°

(C) 60° (D) 30°

Family of Planes

Q21. The equation of the plane through the intersection of the planes
$x+2y+3z-4=0, 4x+3y+2z+1=0$ and passing through the origin will be :

(A) $x+y+z=0$

(B) $7x+4y+z=0$

(C) $17x+14y+11z=0$

(D) $17x+14y+z=0$

Straight Lines

Q22. The equation of straight line passing through the point (a, b, c) and (a, –b, b – c, c – a) is:

(A) $\frac{x-a}{a}=\frac{y-b}{b}=\frac{z-c}{c}$

(B) $\frac{x-a}{2a-b}=\frac{y-b}{2b-c}=\frac{z-c}{2c-a}$

(C) $\frac{x-a}{b}=\frac{y-b}{c}=\frac{z-c}{a}$

(D) $\frac{x-a}{a-b}=\frac{y-b}{b-c}=\frac{z-c}{c-a}$

Q23. The equation of straight line passing through the point (a, b, c) and parallel to z-axis, is :

(A) $\frac{x-a}{0}=\frac{y-b}{1}=\frac{z-c}{1}$

(B) $\frac{x-a}{1}=\frac{y-b}{0}=\frac{z-c}{0}$

(C) $\frac{x-a}{0}=\frac{y-b}{0}=\frac{z-c}{1}$

(D) $\frac{x-a}{1}=\frac{y-b}{1}=\frac{z-c}{0}$

Problems Based on Straight lines

Q24. The angle between two lines
$\frac{x+1}{2}=\frac{y+3}{2}=\frac{z-4}{-1}$ and
$\frac{x-4}{1}=\frac{y+4}{2}=\frac{z+1}{2}$ is :

(A) $\cos^{-1}\left(\frac{1}{9}\right)$ (B) $\cos^{-1}\left(\frac{4}{9}\right)$

(C) $\cos^{-1}\left(\frac{2}{9}\right)$ (D) $\cos^{-1}\left(\frac{3}{9}\right)$

Q25. The cosine of the angle between any two main diagonals of a cube is :

(A) $\frac{1}{3}$ (B) $\frac{1}{\sqrt{3}}$

(C) $\frac{1}{2}$ (D) $\frac{2}{3}$

Q26. The length of the perpendicular from the point (1, 2, 3) to the line
$\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$ will be:

(A) 7 (B) 4

(C) 6 (D) 5

Q27. The co-ordinates of the point where the line through (3, 4, 1) and (5, 1, 6) crosses the *xy*-plane are:

(A) $\left(\frac{11}{5},0,0\right)$ (B) $\left(\frac{13}{5},\frac{21}{5},0\right)$

(C) $\left(\frac{13}{5},\frac{23}{5},0\right)$ (D) $\left(\frac{3}{5},\frac{13}{5},\frac{23}{5}\right)$

Q28. If the lines $\frac{x-1}{-3}=\frac{y-2}{2k}=\frac{z-3}{2}$ and $\frac{x-1}{3k}=\frac{y-5}{1}=\frac{z-6}{-5}$ are at right angles, then the value of k will be :

(A) –10 (B) –7

(C) $-\frac{10}{7}$ (D) $-\frac{7}{10}$

Q29. The acute angle between the line joining the points (2, 1, –3), (–3, 1, 7) and a line parallel to $\frac{x-1}{3}=\frac{y}{4}=\frac{z+3}{5}$ through the point (–1, 0, 4) is :

(A) $\cos^{-1}\left(\frac{7}{5\sqrt{10}}\right)$ (B) $\cos^{-1}\left(\frac{3}{5\sqrt{10}}\right)$

(C) $\cos^{-1}\left(\frac{1}{\sqrt{10}}\right)$ (D) $\cos^{-1}\left(\frac{1}{5\sqrt{10}}\right)$

Q30. The line passing through the points (5, 1, a) and (3, b, 1) crosses the yz-plane at the point $\left(0,\frac{17}{2},\frac{-13}{2}\right)$. Then-

(A) a = 2, b = 8 (B) a = 4, b = 6

(C) a = 6, b = 4 (D) a = 8, b = 2

Q31. The value of 'a' for which the lines $\frac{x-2}{1}=\frac{y-9}{2}=\frac{z-13}{3}$ and $\frac{x-a}{-1}=\frac{y-7}{2}=\frac{z+2}{-3}$ intersect, is :

(A) – 5 (B) – 2

(C) 5 (D) – 3

Q32. For the line $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z-3}{3}$, which one of the following is incorrect?

(A) it lies in the plane $x-2y+z=0$

(B) it is same as line $\frac{x}{1}=\frac{y}{2}=\frac{z}{3}$

(C) it passes through (2, 3, 5)

(D) it is parallel to the plane $x-2y+z-6=0$

Angle Between a Line and a Plane

Q33. If line $\frac{x-x_1}{l}=\frac{y-y_1}{m}=\frac{z-z_1}{n}$ is parallel to the plane $ax+by+cz+d=0$, then :

(A) $\frac{a}{l}+\frac{b}{m}+\frac{c}{n}=0$

(B) $\frac{a}{l}=\frac{b}{m}=\frac{c}{n}$

(C) $al + bm + cn = 0$

(D) None of the above

Q34. The angle between the line $\vec{r}=(\hat{i}+2\hat{j}-\hat{k})+\lambda(\hat{i}-\hat{j}+\hat{k})$ and the normal to the plane $\vec{r}\times(2\hat{i}-\hat{j}+\hat{k})=4$ is :

(A) $\sin^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ (B) $\cos^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

(C) $\tan^{-1}\left(\frac{2\sqrt{2}}{3}\right)$ (D) $\cot^{-1}\left(\frac{2\sqrt{2}}{3}\right)$

Q35. If a plane passes through the point (1, 1, 1) and is perpendicular to the line $\frac{x-1}{3}=\frac{y-1}{0}=\frac{z-1}{4}$, then its perpendicular distance from the origin is :

(A) 4/3 (B) 7/5

(C) 3/4 (D) 1

Q36. The co-ordinates of the point where the line $\frac{x-6}{-1}=\frac{y+1}{0}=\frac{z+3}{4}$, meets the plane $x+y-z=3$ are :

(A) (5, –1, 1) (B) (2, 1, 0)

(C) (1, 2, –6) (D) (7, –1, 7)

Q37. If the angle θ between the line $\frac{x+1}{1}=\frac{y-1}{2}=\frac{z-2}{2}$ and the plane $2x-y+\sqrt{\lambda}\,z+4=0$ is such that $\sin\theta=\frac{1}{3}$ the value of λ is-

(A) $\frac{5}{3}$ (B) $\frac{-3}{5}$

(C) $\frac{3}{4}$ (D) $\frac{-4}{3}$

Shortest Distance Between Two Lines

Q38. If the straight lines $\frac{x-1}{k}=\frac{y-2}{2}=\frac{z-3}{3}$ and $\frac{x-2}{3}=\frac{y-3}{k}=\frac{z-1}{2}$ intersect at a point, then the integral value of k is equal to-

(A) –5 (B) 5 (C) 2 (D) –2

Unsymmetrical Form of Straight Line

Q39. The direction ratios of the line $x-y+z-5=0; x-3y-6=0$ are :

(A) 3, 1, –2

(B) 2, –4, 1

(C) $\frac{3}{\sqrt{14}},\frac{1}{\sqrt{14}},\frac{-2}{\sqrt{14}}$

(D) $\frac{2}{\sqrt{14}},\frac{-4}{\sqrt{14}},\frac{1}{\sqrt{14}}$

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. Consider the plane $\vec{r}.\vec{n}_1=d_1$ and $\vec{r}.\vec{n}_2=d_2$, then which of the follwoing are true -

(A) they are perpendicular if $\vec{n}_1.\vec{n}_2=0$

(B) angle between them is $\cos^{-1}\left(\frac{\vec{n}_1.\vec{n}_2}{|\vec{n}_1||\vec{n}_2|}\right)$

(C) normal form of the equation of plane are $\vec{r}.\vec{n}_1=\frac{d_1}{|\vec{n}_1|}$ & $\vec{r}.\vec{n}_2=\frac{d_2}{|\vec{n}_2|}$

(D) none of these

Q2. The shortest distance between the lines $\frac{x-1}{2}=\frac{y-2}{3}=\frac{z-3}{4}$ and $\frac{x-2}{3}=\frac{y-4}{4}=\frac{z-5}{5}$ is -

(A) $\frac{1}{\sqrt{6}}$ (B) $\frac{1}{6}$

(C) $\frac{1}{3}$ (D) $\frac{1}{\sqrt{3}}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 3 to 4)

Let $L_1 : 2\hat{i} - 3\hat{j} + 4\hat{k} + \lambda(2\hat{i} + 3\hat{j} + 6\hat{k})$

and $L_2 : \hat{i} - \hat{j} + \hat{k} + \mu(2\hat{i} + 3\hat{j} + 6\hat{k})$

Q3. Distance between these lines is

(A) $\frac{\sqrt{10}}{7}$ (B) $\sqrt{10}$ (C) $\frac{10}{7}$ (D) $7\sqrt{10}$

Q4. Equation of the plane containing the lines L_1 and L_2 is

(A) $3x - z - 2 = 0$ (B) $2x + y - z = 0$

(C) $3x - 2y - 3z = 0$ (D) none

Paragraph # 2 (Ques. 5 to 7)

Let P denotes the plane consisting of all points that are equidistant from the points A(– 4, 2, 1) and B(2, – 4, 3) and Q be the plane, $x - y + cz = 1$ where $c \in R$.

Q5. The plane P is parallel to plane Q

(A) for no value of c

(B) if $c = 3$

(C) if $c = 1/3$

(D) if $c = 1$

Q6. If the angle between the planes P and Q is 45° then the product of all possible values of 'c' is

(A) –17 (B) –2

(C) 17 (D) $\frac{24}{17}$

Q7. If the line L with equation

$$\frac{x-1}{1} = \frac{y+2}{3} = \frac{z-7}{-1}$$

intersects the plane P at the point $R(x_0, y_0, z_0)$ then the sum $(x_0 + y_0 + z_0)$ has the value equal to

(A) 12 (B) –15

(C) 13 (D) –11

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C, D, E, F, G and H while the statements in **Column-II** are labelled as (P), (Q), (R), (S), (T), (U), (V) and (W). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q8. If A(0, 1, 0), B(0, 0, 0), C(1, 0, 1) are the vertices of a ΔABC.

Match the entries of column-I with column-II.

Column-I	**Column-II**
(A) Orthocentre of ΔABC	(P) $\frac{\sqrt{2}}{2}$
(B) Circumcentre of ΔABC	(Q) $\frac{\sqrt{3}}{2}$
(C) Area (ΔABC)	(R) $\frac{\sqrt{3}}{3}$
(D) Distance between orthocentre and centroid	(S) $\frac{\sqrt{3}}{6}$

(E) Distance between orthocentre and circumcentre — (T) $(0, 0, 0)$

(F) Distance between circumcentre and centroid — (U) $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$

(G) Incentre of ΔABC — (V) $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

(H) Centroid of ΔABC — (W) $\left(\frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{\sqrt{2}}{\sqrt{1}+\sqrt{2}+\sqrt{3}}, \frac{1}{\sqrt{1}+\sqrt{2}+\sqrt{3}}\right)$

Q9. Consider the following four pairs of lines in column–I and match them with one or more entries in column–II.

Column-I	**Column-II**
(A) $L_1 : x = 1 + t, y = t, z = 2 - 5t$ $L_2 : \vec{r} = (2, 1, -3) + \lambda(2, 2, -10)$	(P) non coplanar lines
(B) $L_1 : \frac{x-1}{2} = \frac{y-3}{2} = \frac{z-2}{-1}$ $L_2 : \frac{x-2}{1} = \frac{y-6}{-1} = \frac{z+2}{3}$	(Q) lines lie in a unique plane
(C) $L_1 : x = -6t, y = 1 + 9t, z = -3t$ $L_2 : x = 1 + 2s, y = 4 - 3s, z = s$	(R) infinite planes containing both the lines
(D) $L_1 : \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ $L_2 : \frac{x-3}{-4} = \frac{y-2}{-3} = \frac{z-1}{2}$	(S) lines are not intersecting

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q10. If the ratio in which the plane $x - 2y + 3z = 17$ divides the line joining the points $(-2, 4, 7)$ and $(3, -5, 8)$ is $m : n$, then find the value of $(m + n)$.

Q11. If the distance of the point $2\hat{i} + \hat{j} - \hat{k}$ from the plane $\vec{r}.(\hat{i} - 2\hat{j} + 4\hat{k}) = 9$ is $\frac{p}{\sqrt{q}}$, then find the value of $(q - p)$.

SUBJECTIVE TYPE

Q12. A vector $\overline{OP}$ is inclined to OX at 45° and OY at 60°. Find the angle at which $\overline{OP}$ is inclined to OZ.

Q13. Find the vector equation of the plane passing through the points $\hat{i}+\hat{j}-2\hat{k}$, $2\hat{i}-\hat{j}+\hat{k}$ and $\hat{i}+2\hat{j}+\hat{k}$.

Q14. Find the vector equation of a plane which is at a distance of 8 units from the origin and which is normal to the vector $2\hat{i}+\hat{j}+2\hat{k}$.

Q15. Find the equations of the bisector planes of the angles between the planes :
$2x-y+2z+3=0$ and $3x-2y+6z+8=0$.

Q16. Find the equation of a plane through the intersection of the planes $\vec{r}.(\hat{i}+3\hat{j}-\hat{k})=5$ and $\vec{r}.(2\hat{i}-\hat{j}+\hat{k})=3$ and passing through the point (2, 1, –2).

Q17. Find the equation of the plane which is perpendicular to the plane $5x+3y+6z+8=0$ and which contains the line of intersection of the planes $x+2y+3z-4=0$ and $2x+y-z+5=0$.

Q18. Find the foot of the perpendicular drawn from the points A(1, 0, 3) to the join of the points B(4, 7, 1) and C(3, 5, 3).

Q19. Find the length of the perpendicular from the point (1, 2, 3) to the line
$\frac{x-6}{3}=\frac{y-7}{2}=\frac{z-7}{-2}$.

Q20. Find the image of the point (1, 6, 3) in the line
$\frac{x}{1}=\frac{y-1}{2}=\frac{z-2}{3}$.

Q21. Find the image of the point (1, 2, 3) in the line $\vec{r}=(6\hat{i}+7\hat{j}+7\hat{k})+\lambda(3\hat{i}+2\hat{j}-2\hat{k})$.

Q22. Find the distance between the point with position vector $-\hat{i}-5\hat{j}-10\hat{k}$ and the point of intersection of the line $\frac{x-2}{3}=\frac{y+1}{4}=\frac{z-2}{12}$ with the plane $x-y+z=5$.

Q23. Show that the lines $\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(3\hat{i}-\hat{j})$ and $\vec{r}=(4\hat{i}-k)+\mu(2\hat{i}+3\hat{k})$ are coplanar. Also, find the plane containing these two lines.

Q24. Prove that the lines $\frac{x+1}{3}=\frac{y+3}{5}=\frac{x+5}{7}$ and $\frac{x-2}{1}=\frac{y-4}{4}=\frac{z-6}{7}$ are coplanar. Also, find the plane containing these two lines.

Q25. Show that the lines $\frac{x-1}{3}=\frac{y+1}{2}=\frac{z-1}{5}$ and $\frac{x+2}{4}=\frac{y-1}{3}=\frac{z+1}{-2}$ do not intersect.

Q26. Show that the lines $\vec{r}=(\hat{i}+\hat{j}-\hat{k})+\lambda(3\hat{i}-\hat{j})$ and $\vec{r}=(4\hat{i}-\hat{k})+\mu(2\hat{i}+3\hat{k})$ intersect. Find their point of intersection.

Q27. Find the shortest distance between the lines
$\vec{r}=(4\hat{i}-\hat{j})+\lambda(\hat{i}+2\hat{j}-3\hat{k})$
and $\vec{r}=(\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}+4\hat{j}-5\hat{k})$.

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The equation of the plane passing through the points $(2,-1,0),(3,-4,5)$ and parallel to the line $2x = 3y = 4z$ is

(A) $125x - 90y - 79z = 340$

(B) $32x - 21y - 36z = 85$

(C) $73x + 61y - 22z = 85$

(D) $29x - 27y - 22z = 85$

Q2. Vector equation of the plane $\vec{r} = \hat{i} - \hat{j} + \lambda(\hat{i} + \hat{j} + \hat{k}) + \mu(\hat{i} - 2\hat{j} + 3\hat{k})$ in the scalar dot product form is -

(A) $\vec{r}.(5\hat{i} - 2\hat{j} + 3\hat{k}) = 7$

(B) $\vec{r}.(5\hat{i} + 2\hat{j} - 3\hat{k}) = 7$

(C) $\vec{r}.(5\hat{i} - 2\hat{j} - 3\hat{k}) = 7$

(D) $\vec{r}.(5\hat{i} + 2\hat{j} + 3\hat{k}) = 7$

Q3. The intercept made by the plane $\vec{r}.\vec{n} = q$ on the x-axis is -

(A) $\dfrac{q}{\hat{i}.\vec{n}}$ (B) $\dfrac{\hat{i}.\vec{n}}{q}$

(C) $(\hat{i}.\vec{n})q$ (D) $\dfrac{q}{|\vec{n}|}$

Q4. If the plane $xbc + yac + zab = abc$ cuts x, y & z-axis in A, B & C respectively then area of ΔABC is -

(A) $\dfrac{1}{2}\sqrt{a^2b^2 + b^2c^2 + c^2a^2}$

(B) $\dfrac{1}{2}\sqrt{ab + bc + ca}$

(C) $\dfrac{1}{2}\sqrt{a^2 + b^2 + c^2}$

(D) $\dfrac{1}{2}\sqrt{\dfrac{a^3 + b^3 + c^3}{a + b + c}}$

Q5. If from the point P(f, g, h) perpendiculars PL, PM be drawn to yz and zx planes then the equation to the plane OLM is -

(A) $\dfrac{x}{f} + \dfrac{y}{g} + \dfrac{z}{h} = 0$ (B) $\dfrac{x}{f} + \dfrac{y}{g} - \dfrac{z}{h} = 0$

(C) $\dfrac{x}{f} - \dfrac{y}{g} + \dfrac{z}{h} = 0$ (D) $-\dfrac{x}{f} + \dfrac{y}{g} + \dfrac{z}{h} = 0$

Q6. A variable plane at a distance of 1 unit from the origin cut the coordinate axis at A, B & C. If centroid of triangle ABC is D(x, y, z) satisfy the relation $\dfrac{1}{x^2} + \dfrac{1}{y^2} + \dfrac{1}{z^2} = k$, then value of k is -

(A) 3 (B) 1

(C) 1/3 (D) 9

Q7. Equation of the plane such that foot of altitude drawn from (–1, 1, 1) to the plane has the coordinate (3, –2, –1), is -

(A) $x + y + z = 0$

(B) $4x - 3y - 2z = 20$

(C) $3x + y - z = 8$

(D) $4x + 3y - z = 7$

Q8. A plane passes through (1, –2, 1) and is perpendicular to two planes $2x - 2y + z = 0$ and $x - y + 2z = 4$. The distance of the plane from the point (1, 2, 3) is -

(A) 0 (B) 1

(C) $\sqrt{2}$ (D) $2\sqrt{2}$

Q9. Consider the plane $(x, y, z) = (0, 1, 1) + \lambda(1, -1, 1) + \mu(2, -1, 0)$. The distance of this plane from the origin is -

(A) 1/3 (B) $\frac{\sqrt{3}}{2}$

(C) $\sqrt{\frac{3}{2}}$ (D) $\frac{2}{\sqrt{3}}$

Q10. Let ABCD be a tetrahedron such that the edges AB, AC and AD are mutually perpendicular. Let the area of triangles ABC, ACD and ADB be 3, 4 and 5 sq. units respectively. Then the area of the triangle BCD, is -

(A) $5\sqrt{2}$ (B) 5

(C) $\frac{5}{\sqrt{2}}$ (D) $\frac{5}{2}$

Q11. A(1, –1, –3), B(2, 1, –2) & C(–5, 2, –6) are the position vectors of the vertices of a triangle ABC. The length of the bisector of its internal angle at A is :

(A) $\sqrt{10}/4$ (B) $3\sqrt{10}/4$

(C) $\sqrt{10}$ (D) none

Q12. The equation of a plane passing through the line of intersection of the planes $x + y + z = 5$, $2x - y + 3z = 1$ and parallel to the line $y = z = 0$ is -

(A) $3x - z = 9$ (B) $3y - z = 9$

(C) $x - 3z = 9$ (D) $y - 3z = 9$

Q13. The distance of the point (–1, 2, 6) from the line $\frac{x-2}{6} = \frac{y-3}{3} = \frac{z+4}{-4}$, is equal to -

(A) 7 units (B) 9 units

(C) 10 units (D) 12 units

Q14. The point of intersection of the lines

$\frac{x+1}{3} = \frac{y+3}{5} = \frac{z+5}{7}$ and

$\frac{x-2}{1} = \frac{y-4}{3} = \frac{z-6}{5}$ is -

(A) $\left(\frac{1}{3}, -\frac{1}{3}, -\frac{2}{3}\right)$ (B) $\left(\frac{1}{2}, -\frac{1}{2}, -\frac{3}{2}\right)$

(C) $\left(\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right)$ (D) $\left(\frac{1}{3}, \frac{1}{3}, \frac{1}{3}\right)$

Q15. If the lines whose direction cosines are given by $a\ell + bm + cn = 0$ and $fmn + gn\ell + h\ell m = 0$ are perpendicular, then $\frac{f}{a} + \frac{g}{b} + \frac{h}{b}$ equals-

(A) 0 (B) –1

(C) 1 (D) None of these

Q16. The line which contains all points (x, y, z) which are of the form $(x, y, z) = (2, -2, 5) + \lambda(1, -3, 2)$ intersects the plane $2x - 3y + 4z = 163$ at P and intersects the YZ plane at Q. If the distance PQ is $a\sqrt{b}$, where $a, b \in N$ and $a > 3$ then $(a + b)$ equals -

(A) 23 (B) 95

(C) 27 (D) none of these

Q17. The distance of the point (1, –2, 3) from the plane $x - y + z - 5 = 0$, Measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-1}{-6}$, is equal to -

(A) 1 unit (B) 2 units

(C) 3 units (D) None of these

Q18. If the plane $2x - 3y + 6z - 11 = 0$ makes an angle $\sin^{-1}(k)$ with x-axis, then k is equal to -

(A) $\frac{\sqrt{3}}{2}$ (B) $\frac{2}{7}$

(C) $\frac{\sqrt{2}}{3}$ (D) 1

Q19. Equation of the plane containing the lines $\frac{x}{1}=\frac{y-2}{3}=\frac{z+4}{-1}$ and $\frac{x-4}{2}=\frac{y}{3}=\frac{z}{1}$ is -

(A) $x+y-4z=6$

(B) $x-y+4z=6$

(C) $x+y+4z=6$

(D) None of these

Q20. The length of projection of the line segment joining the points (1, –1, 0) and (–1, 0, 1), to the plane $2x+y+6z=1$, is equal to

(A) $\sqrt{\frac{256}{41}}$ (B) $\sqrt{\frac{237}{41}}$

(C) $\sqrt{\frac{137}{41}}$ (D) $\sqrt{\frac{155}{41}}$

Q21. Reflection of the line $\frac{x-1}{-1}=\frac{y-2}{3}=\frac{z-4}{1}$ in the plane $x+y+z=7$ is -

(A) $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-4}{1}$

(B) $\frac{x-1}{-3}=\frac{y-2}{-1}=\frac{z-4}{1}$

(C) $\frac{x-1}{-3}=\frac{y-2}{1}=\frac{z-4}{-1}$

(D) $\frac{x-1}{3}=\frac{y-2}{1}=\frac{z-4}{1}$

Q22. Let L_1 be the line $\vec{r}_1=2\hat{i}+\hat{j}-\hat{k}+\lambda(\hat{i}+2\hat{k})$ and let L_2 be the line $\vec{r}_2=3\hat{i}+\hat{j}-\mu(\hat{i}+\hat{j}-\hat{k})$. Let Π be the plane which contains the line L_1 and is parallel to L_2. The distance of the plane Π from the origin is -

(A) $\sqrt{2/7}$ (B) 1/7

(C) $\sqrt{6}$ (D) none of these

Q23. The expression in the vector form for the point $\vec{r}_1$ of intersection of the plane $\vec{r}\cdot\vec{n}=d$ and the perpendicular line $\vec{r}=\vec{r}_0+t\vec{n}$ where t is a parameter given by -

(A) $\vec{r}_1=\vec{r}_0+\left(\frac{d-\vec{r}_0\cdot\vec{n}}{\vec{n}^2}\right)\vec{n}$

(B) $\vec{r}_1=\vec{r}_0-\left(\frac{\vec{r}_0\cdot\vec{n}}{\vec{n}^2}\right)\vec{n}$

(C) $\vec{r}_1=\vec{r}_0-\left(\frac{\vec{r}_0\cdot\vec{n}-d}{|\vec{n}|}\right)\vec{n}$

(D) $\vec{r}_1=\vec{r}_0-\left(\frac{\vec{r}_0\cdot\vec{n}}{|\vec{n}|}\right)\vec{n}$

Q24. The value of m for which straight line $3x-2y+z+3=0=4x-3y+4z+1$ is parallel to the plane $2x-y+mz-2=0$ is

(A) –2 (B) 8

(C) – 18 (D) 11

Q25. A variable plane forms a tetrahedron of constant volume $64K^3$ with the coordinate planes and the origin, then locus of the centroid of the tetrahedron is -

(A) $x^3+y^3+z^3=6K^2$

(B) $xyz=6k^3$

(C) $x^2+y^2+z^2=4K^2$

(D) $x^{-2}+y^{-2}+z^{-2}=4k^{-2}$

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If the line $\vec{r}_1 = 2\hat{i} - \hat{j} + 3\hat{k} + \lambda(\hat{i} + \hat{j} + \sqrt{2}\hat{k})$ makes angles α, β, γ with xy, yz and zx planes respectively then which one of the following are not possible ?

(A) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$ and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

(B) $\tan^2\alpha + \tan^2\beta + \tan^2\gamma = 7$ and $\cot^2\alpha + \cot^2\beta + \cot^2\gamma = 5/3$

(C) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$ and $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) $\sec^2\alpha + \sec^2\beta + \sec^2\gamma = 10$ and $\operatorname{cosec}^2\alpha + \operatorname{cosec}^2\beta + \operatorname{cosec}^2\gamma = 14/3$

Q2. A plane meets the coordinate axes in A, B, C such that the centroid of the triangle ABC is the point $(1, r, r^2)$. The plane passes through the point $(4, -8, 15)$ if 'r' can equal to-

(A) -3 (B) 3

(C) 5 (D) -5

Q3. The points A(5, –1, 1), B(7, –4, 7), C(1, –6, 10) and D(–1, –3, 4) are the vertices of a -

(A) parallelogram (B) rectangle

(C) rhombus (D) square

Q4. If P_1, P_2, P_3 denotes the perpendicular distances of the plane $2x - 3y + 4z + 2 = 0$ from the parallel planes: $2x - 3y + 4z + 6 = 0$, $4x - 6y + 8z + 3 = 0$ and $2x - 3y + 4z - 6 = 0$ respectively, then -

(A) $P_1 + 8P_2 - P_3 = 0$

(B) $P_3 = 16P_2$

(C) $8P_2 = P_1$

(D) $P_1 + 2P_2 + 3P_3 = \sqrt{29}$

Q5. If two pairs of opposite edges of a tetrahedron are perpendicular then -

(A) the third is also perpendicular

(B) the third pair is inclined at 60°

(C) the third pair is inclined at 45°

(D) Options (B) and (C) are false

Q6. The equation of a plane bisecting the angle between the plane $2x - y + 2z + 3 = 0$ and $3x - 2y + 6z + 8 = 0$ is -

(A) $5x - y - 4z - 45 = 0$

(B) $5x - y - 4z - 3 = 0$

(C) $23x - 13y + 32z + 45 = 0$

(D) $23x - 13y + 32z + 5 = 0$

Q7. If ℓ_1, m_1, n_1 and ℓ_2, m_2, n_2 are DCs of the two lines inclined to each other at an angle θ, then the DCs of the bisector of the angle between these lines are -

(A) $\frac{\ell_1 + \ell_2}{2\sin\theta/2}, \frac{m_1 + m_2}{2\sin\theta/2}, \frac{n_1 + n_2}{2\sin\theta/2}$

(B) $\frac{\ell_1 + \ell_2}{2\cos\theta/2}, \frac{m_1 + m_2}{2\cos\theta/2}, \frac{n_1 + n_2}{2\cos\theta/2}$

(C) $\frac{\ell_1 - \ell_2}{2\sin\theta/2}, \frac{m_1 - m_2}{2\sin\theta/2}, \frac{n_1 - n_2}{2\sin\theta/2}$

(D) $\frac{\ell_1 - \ell_2}{2\cos\theta/2}, \frac{m_1 - m_2}{2\cos\theta/2}, \frac{n_1 - n_2}{2\cos\theta/2}$

Q8. Consider the family of planes $x + y + z = c$ where c is a parameter intersecting the coordinate axes at P, Q, R and α, β, γ are the angles made by each member of this family with positive x, y and z axis. Which of the following interpretations holds good for this family -

(A) each member of this family is equally inclined with the coordinate axes.

(B) $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1$

(C) $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 2$

(D) for $c = 3$ area of the triangle PQR is $3\sqrt{3}$ sq. units.

Q9. A non-zero vector $\vec{a}$ is parallel to the line of

intersection of the plane determined by the vectors $\hat{i}$, $\hat{i}+\hat{j}$ and the plane determined by the vectors $\hat{i}-\hat{j}$, $\hat{i}-\hat{k}$. The possible angle between $\vec{a}$ and $\hat{i}-2\hat{j}+2\hat{k}$ is -

(A) $\pi/3$ (B) $\pi/4$

(C) $\pi/6$ (D) $3\pi/4$

Q10. A line L passing through the point P(1,4,3), is perpendicular to both the lines

$$\frac{x-1}{2}=\frac{y+3}{1}=\frac{z-2}{4}$$

and $\frac{x+2}{3}=\frac{y-4}{2}=\frac{z+1}{-2}$.

If the position vector of point Q on L is (a_1,a_2,a_3) such that $(PQ)^2 = 357$, then $(a_1 + a_2 + a_3)$ can be-

(A) 16 (B) 15

(C) 2 (D) 1

Q11. Points that lie on the lines bisecting the angle between the lines $\frac{x-2}{2}=\frac{y-3}{3}=\frac{z-6}{6}$ and

$\frac{x-2}{3}=\frac{y-3}{6}=\frac{z-6}{2}$ are -

(A) (7, 12, 14)

(B) (0, –3, 14)

(C) (1, 0, 10)

(D) (–3, –6, –2)

Q12. The coordinates of a point on the line $\frac{x-1}{2}=\frac{y+1}{-3}=z$ at a distance $4\sqrt{14}$ from the point (1, –1, 0) are -

(A) (9, –13, 4)

(B) $(8\sqrt{14}+1,-12\sqrt{14}-1,4\sqrt{14})$

(C) (–7, 11, –4)

(D) $(-8\sqrt{14}+1,12\sqrt{14}-1,-4\sqrt{14})$

Q13. The lines $\frac{x-2}{1}=\frac{y-3}{1}=\frac{z-4}{-k}$ and

$\frac{x-1}{k}=\frac{y-4}{2}=\frac{z-5}{1}$ are coplanar.

If:

(A) k = 0 (B) k = –1

(C) k = –3 (D) k = 3

Q14. The plane x – 2y + 7z +21 = 0

(A) contains the line $\frac{x+1}{-3}=\frac{y-3}{2}=\frac{z+2}{1}$

(B) contains the point (0, 7, –1)

(C) is perpendicular to the line $\frac{x}{1}=\frac{y}{-2}=\frac{z}{7}$

(D) is parallel to the line $\frac{x}{1}=\frac{y}{-2}=\frac{z}{7}$

Q15. The lines $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-3}{\lambda}$ and

$\frac{x}{1}=\frac{y}{2}=\frac{z+1}{-1}$ are

(A) coplanar for all λ

(B) coplanar for $\lambda = 19/3$

(C) if coplanar then intersect at

$$\left(-\frac{1}{5},-\frac{2}{5},-\frac{4}{5}\right)$$

(D) intersect at $\left(\frac{1}{2},-\frac{1}{2},-1\right)$

Q16. The plane containing the lines $\vec{r}=\vec{a}+t\vec{a}'$ and $\vec{r}=\vec{a}'+s\vec{a}$

(A) must be parallel to $\vec{a}\times\vec{a}'$

(B) must be the perpendicular to $\vec{a}\times\vec{a}'$

(C) must be $[\vec{r}, \vec{a}, \vec{a}'] = 0$

(D) $(\vec{r}-\vec{a}).(\vec{a}\times\vec{a}') = 0$

Q17. Let $6x + 4y - 5z = 4$, $x - 5y + 2z = 12$ and $\frac{x-9}{2} = \frac{y+4}{-1} = \frac{z-5}{1}$ be two lines then -

(A) the angle between them must be $\frac{\pi}{3}$

(B) the angle between them must be $\cos^{-1}\frac{5}{6}$

(C) the plane containing them must be $x + y - z = 0$

(D) they are non-coplanar

Q18. Given the equations of the line $3x - y + z + 1 = 0$, $5x + y + 3z = 0$.
Then which of the following is correct ?

(A) Symmetrical form of the equations of line is $\frac{x}{2} = \frac{y-\frac{1}{8}}{-1} = \frac{z+\frac{5}{8}}{1}$

(B) symmetrical form of the equations of line is $\frac{x+\frac{1}{8}}{1} = \frac{y-\frac{5}{8}}{1} = \frac{z}{-2}$

(C) Eqn. of the plane through (2, 1, 4) and prependicular to the given lines is $2x - y + z - 7 = 0$.

(D) Eqn. of the plane through (2, 1, 4) and prependicular to the given lines is $x + y - 2z + 5 = 0$.

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 19 to 21)

Given four points A(2, 1, 0); B(1, 0, 1); C(3, 0, 1) and D(0, 0, 2). The point D lies on a line L orthogonal to the plane determined by the point A, B, and C.

Q19. Equation of the plane ABC is -

(A) $x + y + z - 3 = 0$

(B) $y + z - 1 = 0$

(C) $x + z - 1 = 0$

(D) $2y + z - 1 = 0$

Q20. Equation of the line L is -

(A) $\vec{r} = 2\hat{k} + \lambda(\hat{i} + \hat{k})$

(B) $\vec{r} = 2\hat{k} + \lambda(2\hat{j} + \hat{k})$

(C) $\vec{r} = 2\hat{k} + \lambda(\hat{j} + \hat{k})$

(D) none of these

Q21. Perpendicular distance of D from the plane ABC, is -

(A) $\sqrt{2}$ (B) $\frac{1}{2}$

(C) 2 (D) $\frac{1}{\sqrt{2}}$

Paragraph # 2 (Ques. 22 to 24)

If a line passes through $P(x_1, y_1, z_1)$ and having Dr's a, b, c, then the equation of line is

$$\frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c}$$

and equation of plane per-pendicular to it and passing through P is $a(x - x_1) + b(y - y_1) + c(z - z_1) = 0$.

Further equation of plane through the intersection of the two planes:

$a_1x + b_1y + c_1z + d_1 = 0$

and $a_2x + b_2y + c_2z + d_2 = 0$ is

$(a_1x + b_1y + c_1z + d_1) + k(a_2x + b_2y + c_2z + d_2) = 0$

Q22. The distance of the point (1, –2, 3) from the plane $x - y + z = 5$ measured parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z-3}{-4}$ is -

(A) $\frac{1}{5}\sqrt{21}$ (B) $\frac{1}{5}\sqrt{29}$

(C) $\frac{1}{5}\sqrt{13}$ (D) $\frac{2}{\sqrt{5}}$

Q23. The equation of the plane through (0, 2, 4) and containing the line $\frac{x+3}{3}=\frac{y-1}{4}=\frac{z-2}{-2}$ is -

(A) $x-2y+4z-12=0$

(B) $5x+y+9z-38=0$

(C) $10x-12y-9z+60=0$

(D) $7x+5y-3z+2=0$

Q24. The plane $x-y-z=2$ is rotated through 90° about its line of intersection with the plane $x+2y+z=2$. Then equation of this plane in new position is -

(A) $5x+4y+z-10=0$

(B) $4x+5y+3z=0$

(C) $2x+y+2z=9$

(D) $3x+4y-5z=9$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q25. P(0, 3, – 2); Q(3, 7, – 1) and R(1, – 3, – 1) are 3 given points. Let L_1 be the line passing through P and Q and L_2 be the line through R and parallel to the vector $\vec{V}=\hat{i}+\hat{k}$.

Column-I	**Column-II**
(A) perpendicular distance of P from L_2	(P) $7\sqrt{3}$
(B) shortest distance between L_1 and L_2	(Q) 2
(C) area of the triangle PQR	(R) 6
(D) distance from (0, 0, 0) to the plane PQR	(S) $\frac{19}{\sqrt{147}}$

Q26. Match the following pair of planes with their lines of intersections :

Column-I	**Column-II**
(A) $x+y=0=y+z$	(P) $\frac{x-2}{0}=\frac{y-2007}{-1}=\frac{z+2004}{1}$
(B) $x=2, y=3$	(Q) $\frac{x-2}{0}=\frac{y}{-1}=\frac{z-1}{1}$
(C) $x=2, y+z=3$	(R) $x=-y=z$
(D) $x=2, x+y+z=3$	(S) $\frac{x-2}{0}=\frac{y-3}{0}=\frac{z}{1}$

Q27. Consider three planes : $P_1 \equiv 2x + y + z = 1$

$P_2 \equiv x - y + z = 2$

$P_3 \equiv \alpha x - y + 3z = 5$

The three planes intersects each other at point P on XOY plane and at point Q on YOZ plane. O is the origin.

Column-I	Column-II
(A) The value of α is	(P) 1
(B) The length of projection of PQ on x-axis is	
(C) If the co-ordinates of point R situated at a minimum distance from point 'O' on the line PQ are (a, b, c), then value of $7a + 14b + 14c$ is	(Q) 2
	(R) 4
(D) If the area of ΔPOQ is $\sqrt{\frac{a}{b}}$, then value of $a - b$ is	(S) 3

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q28. Faces ABC and BCD of a tetrahedron ABCD meet at an angle of 30°. The area of face ABC is 120 and the area of face BCD is 80 and BC = 10, then the volume of tetrahedron is ?

Q29. The position vectors of the four angular points of a tetrahedron OABC are (0, 0, 0); (0, 0, 2); (0, 4, 0) and (6, 0, 0) respectively. A point P inside the tetrahedron is at the same distance 'r' from the four plane faces of the tetrahedron. Find the value of 36r.

Q30. Let the equation of the plane containing the line $x - y - z - 4 = 0 = x + y + 2z - 4$ and is parallel to the line of intersection of the planes $2x + 3y + z = 1$ and $x + 3y + 2z = 2$ be $x + Ay + Bz + C = 0$ Compute the value of $|A + B + C|$.

Q31. The equation of the plane which has the property that the point Q(5,4,5) is the reflection of point P(1,2,3) through that plane, is $ax + by + cz = d$ where $a,b,c,d \in N$. Find the least value of $(a + b + c + d)$.

SUBJECTIVE TYPE

Q32. The plane $\ell x + my = 0$ is rotated about its line of intersection with the plane $z = 0$ through an angleθ. Prove that the equation to the plane in new position is $\ell x + my \pm z\sqrt{\ell^2 + m^2}\tan\theta = 0$.

Q33. Find the equations of the two lines through the origin which intersect the line $\frac{x-3}{2} = \frac{y-3}{1} = \frac{z}{1}$ at an angle of $\frac{\pi}{3}$.

Q34. Find the equations to the line which can be drawn from the point (2, –1, 3) per-pendicular to the lines $\frac{x-1}{2} = \frac{y-2}{3} = \frac{z-3}{4}$ and $\frac{x-4}{4} = \frac{y}{5} = \frac{z+3}{3}$ at right angles.

Q35. Find the equation of the plane containing the line $2x - y + z - 3 = 0, 3x + y + z = 5$ and at a distance of $\frac{1}{\sqrt{6}}$ from the point (2, 1, –1).

Q36. P is any point on the plane $\ell x + my + nz = p$. A point Q taken on the line OP (where O is the origin) such that $OP.OQ = p^2$.

Show that the locus of Q is $p(\ell x + my + nz) = x^2 + y^2 + z^2$.

Answer Key

Ex-1									
1	2	3	4	5	6	7	8	9	10
D	C	D	C	C	C	C	A	B	D
11	12	13	14	15	16	17	18	19	20
C	D	A	B	B	D	D	B	A	C
21	22	23	24	25	26	27	28	29	30
C	C	C	B	A	A	C	C	A	C
31	32	33	34	35	36	37	38	39	
D	C	C	B	B	A	A	A	A	

Ex - 1A

1. A,B **2.** A **3.** B **4.** A **5.** C

6. B **7.** A

8. (A) → (T); (B) → (U); (C) → (P); (D) → (R); (E) → (Q); (F) → (S);(G) → (W); (H) → (V)

9. (A) → (R); (B) → (Q); (C) → (Q,S); (D) → (P,S)

10. 13 **11.** 8 **12.** 60° or 120°

13. $\vec{r}.(9\hat{i}+3\hat{j}-\hat{k})=14$ **14.** $\vec{r}.(2\hat{i}+3j+2\hat{k})=\pm(24)$

15. $5x - y - 4z - 3 = 0$ and $23x - 13y + 32z + 45 = 0$

16. $\vec{r}.(3\hat{i}+2\hat{j}+0\hat{k})=8$ **17.** $51x + 15y - 50z + 173 = 0$

18. $\left(\frac{5}{3},\frac{7}{3},\frac{17}{3}\right)$ **19.** 7 units

20. 1,0,7 **21.** 5,8,15

22. Point of intersection: (2, –1, 2) ; Distance: 13 units

23. $\vec{r}.(3\hat{i}+9\hat{j}-2\hat{k})=14s$ **24.** $x - 2y + z = 0$

26. Point of intersection: (4, 0, –1) **27.** $\frac{6}{\sqrt{5}}$ units

Ex-2									
1	2	3	4	5	6	7	8	9	10
D	C	A	A	B	D	B	D	C	A
11	12	13	14	15	16	17	18	19	20
B	B	A	B	A	A	A	B	D	B
21	22	23	24	25					
C	A	A	A	B					

Ex - 2A

1. A,B,D **2.** B,C **3.** A,C **4.** A,B,C,D **5.** A,D

6. B,C **7.** B,C **8.** A,B,C **9.** B,D **10.** B,D

11. A,D **12.** A,C **13.** A,C **14.** A,B,C **15.** B,C

16. B,C,D **17.** A,C **18.** B,D **19.** B **20.** C

21. D **22.** B **23.** C **24.** A

25. $A \rightarrow R$; $B \rightarrow Q$; $C \rightarrow P$; $D \rightarrow S$

26. $A \rightarrow R$; $B \rightarrow S$; $C \rightarrow P$; $D \rightarrow Q$

27. $A \rightarrow R$; $B \rightarrow P$; $C \rightarrow Q$; $D \rightarrow S$

28. 320 **29.** 24 **30.** 11 **31.** 17

33. $\frac{x}{1}=\frac{y}{2}=\frac{z}{-1}$ or $\frac{x}{-1}=\frac{y}{1}=\frac{z}{-2}$

34. $\frac{x-2}{11}=\frac{y+1}{-10}=\frac{z-3}{2}$

35. Equation of planes are : $2x - y + z - 3 = 0$ and $62x + 29y + 19z - 105 = 0$

-: Important Notes & Formulas :-

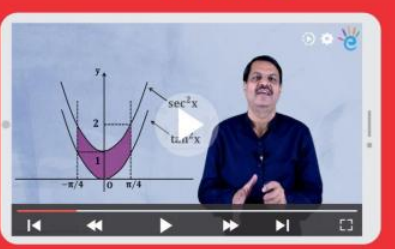
Get Personalised Learning for

IIT-JEE/NEET &

Class 9,10,NTSE,Olympiad *With*

eSaral APP

Video Lectures Covering Entire Syllabus

Physics	Chemistry	Maths
		
Saransh Sir	**Prateek Sir**	**N.K.Gupta Sir**
IIT Bombay (AIR 41)	**IIT Bombay**	**IIT Kanpur**

Offline Course Available On Tablet Without Internet

- Solved Study Material
- Doubt Solving
- Online Test + Analysis
- Mentorship

Download FREE eSaral App

GET IT ON Google Play

+91-6376 440 597

www.esaral.com

eSaral है, तो सब सरल है।